COUNTABLE STRONGLY CHOQUET–DENY GROUPS ARE PRECISELY THE VIRTUALLY FC GROUPS

Abstract. We show that the countable strongly Choquet–Deny groups are precisely the virtually FC groups, extending the finitely generated classification of Dantas and Doucha. The proof uses their criterion for invariant strictly convex renormability and the analytic construction in their Heisenberg example. The positive direction replaces commutativity by finite conjugacy averaging. For the converse, we construct the sequence required by their obstruction whenever the FC-center has infinite index. The same characterization holds for their positive strongly Choquet–Deny condition.

1. Introduction

Dantas and Doucha introduced strongly Choquet–Deny groups and proved that a countable group has this property if and only if every uniformly bounded representation on a separable real Banach space admits an equivalent invariant strictly convex norm [DD, Theorem 4.2]. Their work follows the study of invariant renormings in [D]; the connection between harmonic vectors and strict convexity also appears in Jaworski–Neufang [JN, Proposition 2.5]. Dantas–Doucha classified the finitely generated strongly Choquet–Deny groups as the virtually abelian groups [DD, Theorem 5.6]. This note extends their classification to arbitrary countable groups.

Recall that a group is *FC* if all its conjugacy classes are finite, and *virtually FC* if it has an FC subgroup of finite index. Its FC-center $\mathrm{FC}(G)$ is the subgroup of elements with finite conjugacy class in G. The strong Choquet–Deny conditions in the following statement are those of [DD, Definition 2.4], recalled in Section 2.

Theorem 1.1. *For a countable group G, the following conditions are equivalent:*

1. *G is virtually FC.*
2. *Every uniformly bounded representation of G on a separable real Banach space admits an equivalent G-invariant strictly convex norm.*
3. *G is strongly Choquet–Deny.*
4. *G is positively strongly Choquet–Deny.*

A finitely generated FC group has a finite-index center: intersect the centralizers of a finite generating set. Since a finite-index subgroup of a finitely generated group is finitely generated, virtual FC and virtual abelianness coincide in that setting. They differ for general countable groups; for example, $\bigoplus_{n\geq 1} S_3$ is FC but not virtually abelian (Example 6.2).

1.1. Proof and relation to earlier work.

We use two extensions of the methods of [DD]. For the positive direction, finite conjugacy averages show that a midpoint decomposition in an absolutely convex orbit hull preserves all norms and linear relations in the cyclic representation. This produces two contractions whose average is the identity, extending the abelian operator argument of [DD, Theorem 5.3]. Kakutani's extremality theorem gives orbit extremality, and the criterion in [DD, Theorem 3.1] supplies the invariant strictly convex norm. Finite-index extension then treats virtually FC groups.

For the converse, [DD, Example 5.5, Claim 2] constructs an obstruction from a sequence satisfying the three product conditions in Claim 1 of that example. Proposition 4.1 records

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this implication, with a minor change to the damping function. The additional group-theoretic argument is Theorem 5.5, which constructs such a sequence whenever $[G : \mathrm{FC}(G)] = \infty$. It is a finite extension argument using centralizer cosets and the equation-avoidance lemma (Lemma 5.3). Its relation to the rewritability theorem of Blyth–Rhemtulla [BR] and the switching constructions of [FTVF, FHTVF, EK] is explained in Section 5.

For comparison, a countable group is *Choquet–Deny* if every bounded μ-harmonic function is constant on left cosets of the subgroup generated by $\mathrm{supp}(\mu)$, for every probability measure μ [DD, Definition 2.2]. These are precisely the groups with no nontrivial ICC quotient [FHTVF, Theorem 1], or equivalently the hyper-FC-central groups. Here ICC means nontrivial with every nonidentity conjugacy class infinite; hyper-FC-central means that iterating FC-centers in successive quotients, with unions at limit stages, exhausts the group. The positive direction of this ordinary classification goes back to [J, Theorems 4.5 and 4.8]. Theorem 1.1 requires finite index already at the first FC-center. The examples in Section 6 show that this is a strictly stronger requirement.

The equivalence of the positive and signed strong conditions for the same countable group is a consequence of the classification. The earlier relation in [DD, Proposition 2.6] identifies the signed condition for G with the positive condition for $G \times \mathbb{Z}_2$.

2. Preliminaries

All groups are discrete, and all Banach spaces and function spaces are over $\mathbb{R}$. All convex hull closures are taken in norm. A representation $\pi : G \to \mathrm{GL}(X)$ is *uniformly bounded* if $M = \sup_{g\in G} \|\pi(g)\| < \infty$. In this case

$$\|x\|_0 = \sup_{g\in G} \|\pi(g)x\|$$

is an equivalent invariant norm, with $\|x\| \le \|x\|_0 \le M\|x\|$. Thus it suffices to treat isometric representations. A norm N is *strictly convex* if $N((x+y)/2) < 1$ whenever $x \ne y$ and $N(x) = N(y) = 1$.

2.1. Measures and translations.

We follow [DD, Section 2.2] in writing $\mathrm{Prob}(G)$ for the probability measures and $\mathrm{Meas}(G) = B_{\ell^1(G)}$ for the real signed measures of total variation at most one. For $\mu \in \ell^1(G)$ set $\mu^*(h) = \mu(h^{-1})$. Their convolution convention is

$$(f * \mu)(g) = \sum_{h\in G} f(gh)\mu(h^{-1}), \qquad (f * \mu^*)(g) = \sum_{h\in G} \mu(h)f(gh).$$

We also write $(R_h f)(g) = f(gh)$ for right translation. Thus $R_h R_k = R_{hk}$ and $f*\mu^* = \sum_h \mu(h)R_h f$, with convergence in supremum norm. In particular, right convolution preserves the closed span of the right translates of f. A bounded function is μ-harmonic if $f * \mu^* = f$. The symbol δ_g denotes a point mass or the indicator of $\{g\}$, according to the ambient space.

Definition 2.1 ([DD, Definition 2.4])**.** A countable group G is *strongly Choquet–Deny* if the following implication holds. Let $\mu_n, \nu_n, \nu_n' \in \mathrm{Meas}(G)$ with $\mu_n = (\nu_n + \nu_n')/2$, and let $f \in \ell^\infty(G)$ be fixed. If

$$\|f * \mu_n^* - f\|_\infty \longrightarrow 0, \qquad \sum_{n=1}^{\infty} \|d_{n+1} - d_n\|_\infty < \infty, \quad d_n = f * (\nu_n^* - (\nu_n')^*), \tag{1}$$

then $\|d_n\|_\infty \to 0$. The group is *positively strongly Choquet–Deny* if the same implication holds with all measures in $\mathrm{Prob}(G)$.

The summability in (1) makes (d_n) converge in supremum norm, so this is equivalent to the subsequence conclusion in [DD, Definition 2.4]. Restricting the measures shows that the signed condition implies the positive one [DD, Proposition 2.5].

2.2. **Orbit hulls and the FC-center.** Write $\mathrm{Orb}_G(x) = \{\pi(g)x : g \in G\}$ and $\mathrm{Orb}_G(x, -x) = \mathrm{Orb}_G(x) \cup \mathrm{Orb}_G(-x)$, as in [DD, Section 2.1]. Its closed convex hull is the closed absolutely convex hull of $\mathrm{Orb}_G(x)$. We use the following criterion.

Theorem 2.2 (Dantas–Doucha, [DD, Theorem 3.1])**.** *An isometric representation of a group G on a separable real Banach space X admits an equivalent invariant strictly convex norm if and only if every $x \in X$ is an extreme point of $\overline{\mathrm{co}}(\mathrm{Orb}_G(x, -x))$.*

For countable groups, [DD, Theorem 4.2] further identifies the strongly Choquet–Deny property with the existence of such a norm in every separable uniformly bounded representation. We use this equivalence in the proof of Theorem 1.1.

Necessity in Theorem 2.2 holds without separability: all points of this hull have invariant norm at most that of x, so strict convexity excludes a nontrivial midpoint at x. In our converse the endpoints will even lie in $\overline{\mathrm{co}}(\mathrm{Orb}_G(x))$.

For $g \in G$ let $C_G(g) = \{h \in G : hg = gh\}$ be its centralizer. The FC-center is

$$\mathrm{FC}(G) = \{g \in G : [G : C_G(g)] < \infty\}.$$

The index equals the cardinality of the conjugacy class of g. The FC-center is a characteristic subgroup: conjugates of a product belong to the product of the conjugacy classes of its factors, and inversion and automorphisms preserve finiteness of conjugacy classes. See also [EK, Section 5.B]. Moreover,

$$G \text{ is virtually FC} \quad \Longleftrightarrow \quad [G : \mathrm{FC}(G)] < \infty. \tag{2}$$

For the nontrivial implication, let $H \leq G$ be FC of finite index. The G-conjugacy class of $h \in H$ is a finite union of conjugates of its finite H-conjugacy class. Thus $H \subseteq \mathrm{FC}(G)$.

3. Invariant renormings for virtually FC groups

We prove the positive direction of Theorem 1.1. No countability assumption on the group is needed in this section. The proof follows the contractive-operator strategy of [DD, Theorem 5.3]. Finite conjugacy averaging also appears in Jaworski's unitary-representation argument [J, proof of Theorem 4.4]. Here we average squared norms of group-algebra translates to preserve all linear relations in a cyclic Banach representation.

Lemma 3.1 (Kakutani; see [EFHN, Theorem 8.28])**.** *For every real Banach space E, the identity is an extreme point of the closed unit ball of $\mathcal{B}(E)$.*

Proof. We use the geometric proof in [EFHN, Theorem 8.28]. Suppose $A, B : E \to E$ are contractions with $(A + B)/2 = I$. For every extreme point φ of the dual unit ball B_{E^*},

$$\varphi = \frac{A^*\varphi + B^*\varphi}{2}, \qquad A^*\varphi, B^*\varphi \in B_{E^*}.$$

Extremality gives $A^*\varphi = B^*\varphi = \varphi$. By the Kreĭn–Milman theorem, B_{E^*} is the weak-star closed convex hull of its extreme points. Since adjoint operators are weak-star continuous, A^* and B^* fix the whole dual unit ball. Thus $A = B = I$. □

Theorem 3.2 (FC orbit extremality)**.** *Let G be an FC group. In every real isometric Banach representation, without a separability assumption, every vector x is extreme in its closed absolutely convex orbit hull $\overline{\mathrm{co}}(\mathrm{Orb}_G(x, -x))$. Consequently every uniformly bounded representation of G on a separable real Banach space admits an equivalent invariant strictly convex norm.*

Proof. Fix $x = (y+z)/2$ with $y, z \in \overline{\mathrm{co}}(\mathrm{Orb}_G(x, -x))$. For a finitely supported element $a = \sum_g a_g g$ of the real group algebra, write $\pi(a) = \sum_g a_g \pi(g)$ and let

$$\mathcal{O}(a) = \{h^{-1}ah : h \in G\}.$$

This set is finite: the support of a is finite, and every element of that support has finitely many conjugates. Define

$$\Phi_a(v) = \frac{1}{|\mathcal{O}(a)|} \sum_{b \in \mathcal{O}(a)} \|\pi(b)v\|^2.$$

This function is continuous, even, convex, and G-invariant. Indeed, $\pi(b)\pi(h) = \pi(h)\pi(h^{-1}bh)$, the first factor is an isometry, and conjugation permutes $\mathcal{O}(a)$.

Since $y, z \in \overline{\mathrm{co}}(\mathrm{Orb}_G(x, -x))$, continuity, convexity, and invariance give $\Phi_a(y), \Phi_a(z) \le \Phi_a(x)$. Thus

$$\Phi_a(x) \le \frac{\Phi_a(y) + \Phi_a(z)}{2} \le \Phi_a(x).$$

All the individual squared-norm Jensen gaps in this finite average are nonnegative, so they all vanish. Indeed, the triangle inequality gives $\|\pi(b)x\| \le (\|\pi(b)y\| + \|\pi(b)z\|)/2$, and hence for each $b \in \mathcal{O}(a)$,

$$\frac{\|\pi(b)y\|^2 + \|\pi(b)z\|^2}{2} - \|\pi(b)x\|^2 \ge \frac{1}{4}\big(\|\pi(b)y\| - \|\pi(b)z\|\big)^2.$$

In particular, because $a \in \mathcal{O}(a)$,

$$\|\pi(a)y\| = \|\pi(a)z\| = \|\pi(a)x\| \quad \text{for every finitely supported real } a. \tag{3}$$

Put $E = \overline{\mathrm{span}}\{\pi(g)x : g \in G\}$. Both y and z lie in E. On its dense algebraic cyclic subspace define

$$A(\pi(a)x) = \pi(a)y, \qquad B(\pi(a)x) = \pi(a)z.$$

In particular, (3) gives $\pi(a)x = 0 \Rightarrow \pi(a)y = \pi(a)z = 0$. Applying this to the difference of two group-algebra elements shows that the definitions respect every linear relation. The maps are therefore well-defined linear isometries. They extend to contractions $A, B : E \to E$, and $(A+B)/2 = I_E$ because $y + z = 2x$. Lemma 3.1 gives $A = B = I_E$, whence $y = z = x$.

The final assertion follows by first making the representation isometric and then applying Theorem 2.2. □

Corollary 3.3 (Finite-index extension)**.** *Every uniformly bounded representation of a virtually FC group on a separable real Banach space admits an equivalent invariant strictly convex norm.*

Proof. This is the finite-index case of [DD, Lemma 3.2 and Proposition 5.1]; we give a direct proof by averaging. Let $H \le G$ be FC with finite index d, and first make the given representation isometric using $\|\cdot\|_0$. Theorem 3.2 gives an equivalent H-invariant strictly convex norm N_H. Choose representatives $r_1, \dots, r_d$ for the right cosets $H\backslash G$ and set

$$N_G(v) = \frac{1}{d} \sum_{j=1}^{d} N_H(\pi(r_j)v).$$

Right multiplication permutes the right cosets, so the H-invariance of N_H makes N_G invariant under G. Each summand is an equivalent strictly convex norm, and their finite average has the same properties. Normality of H is not required. □

4. The Dantas–Doucha obstruction

We first explain the analytic target of the converse. It is enough to find f and two sequences of probability measures such that

$$f * \nu_n^* \longrightarrow f + \delta_e, \qquad f * (\nu_n')^* \longrightarrow f - \delta_e \tag{4}$$

in supremum norm, with summable errors. Their average makes f approximately harmonic, but their difference has a nonzero limit. Equally, (4) places f at a nontrivial midpoint of its closed convex orbit hull. This is the construction of Dantas–Doucha in [DD, Example 5.5, Claim 2].

4.1. **The three product properties.** We retain the notation of [DD, Example 5.5]. Let $\mathcal{F}$ be the set of finite sequences of distinct positive integers, including the empty sequence $\varnothing$. Write $|v|$ for the length of v, $v(i)$ for its ith entry, and $\mathrm{ran}(v)$ for its set of entries. Given $(s_i)_{i\geq 1} \subseteq G$, put

$$s_v = s_{v(1)} \cdots s_{v(|v|)}, \qquad s_\varnothing = e, \qquad \mathcal{S} = \{s_v : v \in \mathcal{F}\}.$$

Concatenation is denoted by juxtaposition; it belongs to $\mathcal{F}$ only when no entry is repeated. Equalities of sequences mean equality as ordered strings, while equalities of their evaluations hold in G.

The hypotheses below are exactly the three properties in [DD, Example 5.5, Claim 1]. Their roles can be seen before any estimate. Injectivity makes the first entry of a nonempty product well-defined. Appending an unused entry preserves that first entry; appending an entry already used leaves $\mathcal{S}$. Finally, a point outside $\mathcal{S}$ has at most one right successor in $\mathcal{S}$, so averages over large blocks are small there. Thus a sign assigned to the first entry is almost preserved by right averaging, except at the empty sequence, where the first sign is created.

The needed approximation is uniform over all products. A fixed product uses finitely many entries, but products of arbitrarily large length can exhaust an entire averaging block. Give each letter a small positive cost and let w be the total cost of a word. We damp its first-letter sign by $1/(1+w)$. A word that has used many letters from one block has a large cost, and therefore a small value. This is the uniformity mechanism in the proof below.

Proposition 4.1 (Dantas–Doucha obstruction, extracted from [DD, Example 5.5, Claim 2])**.** *Let G be countable and $(s_i)_{i\geq 1} \subseteq G$. Assume that:*

1. *the map $v \mapsto s_v$ on $\mathcal{F}$ is injective;*
2. *if $i \in \mathrm{ran}(v)$, then $s_v s_i \notin \mathcal{S}$;*
3. *if $g \notin \mathcal{S}$, at most one index i satisfies $gs_i \in \mathcal{S}$.*

There are $f \in \ell^\infty(G)$ and $\mu_n, \nu_n, \nu_n' \in \mathrm{Prob}(G)$ with $\mu_n = (\nu_n + \nu_n')/2$ such that

$$\begin{aligned} \|f * \nu_n^* - (f + \delta_e)\|_\infty &\leq 3 \cdot 2^{-n}, \\ \|f * (\nu_n')^* - (f - \delta_e)\|_\infty &\leq 3 \cdot 2^{-n}. \end{aligned} \tag{5}$$

Consequently G is not positively strongly Choquet–Deny. The right-translation representation on the separable cyclic space

$$X_f = \overline{\mathrm{span}}\{R_g f : g \in G\} \subseteq \ell^\infty(G)$$

has no equivalent invariant strictly convex norm; more precisely, $f + \delta_e, f - \delta_e \in \overline{\mathrm{co}}(\mathrm{Orb}_G(f))$.

Proof. We adapt the argument of [DD, Example 5.5, Claim 2]. We retain their blocks, signs, and additive weight, but use reciprocal damping in place of their truncated affine factor. Blocks of size 2^{2n} suffice in place of 2^{2^n}. Partition $\mathbb{N}$ into consecutive blocks N_n with $|N_n| = 2^{2n}$, and put $S_n = \{s_i : i \in N_n\}$. Split S_n into two equally sized sets S_n^+, S_n^-. Let μ_n, ν_n, ν_n' be uniform on S_n, S_n^+, S_n^-, respectively. These are probability measures and $\mu_n = (\nu_n + \nu_n')/2$.

Set $\sigma(s_i) = 1$ on S_n^+ and $\sigma(s_i) = -1$ on S_n^-. Following the notation of [DD], define the additive weight $f' : \mathcal{S} \to [0, \infty)$ by

$$f'(s_v) = \sum_{k \geq 1} 2^{-k} |N_k \cap \mathrm{ran}(v)|.$$

The sum has finitely many nonzero terms, and the first hypothesis makes it well-defined. Define

$$f(s_v) = \frac{\sigma(s_{v(1)})}{1 + f'(s_v)} \quad (v \neq \varnothing), \qquad f(e) = 0, \qquad f(g) = 0 \quad (g \notin \mathcal{S}).$$

Then $\|f\|_\infty = 1$, since $|f(s_i)| = (1 + 2^{-n})^{-1}$ for $s_i \in S_n$. Fix n, and put $\varepsilon = 2^{-n}$ and $L = |S_n^+| = |S_n^-| = 2^{2n-1}$. We estimate the two averages in (5) by the same argument. At e their values are $(1 + \varepsilon)^{-1}$ and $-(1 + \varepsilon)^{-1}$, so each has error at most ε from its target. Outside $\mathcal{S}$, at most one successor contributes, and the error is at most $L^{-1} = 2\varepsilon^2 \leq \varepsilon$.

Let $g = s_v \neq e$, write $w = f'(s_v)$, and consider either half-block. Let r be the number of its entries already used in v. Then $r\varepsilon \leq w$. Each used entry gives a zero successor by the second hypothesis. Every unused entry preserves the first letter and adds ε to the weight. Writing $\sigma = \sigma(s_{v(1)})$, the half-block average is therefore

$$\frac{(1 - r/L)\sigma}{1 + w + \varepsilon}.$$

Its error from $f(g) = \sigma/(1 + w)$ is exactly

$$\frac{\varepsilon}{(1 + w)(1 + w + \varepsilon)} + \frac{r}{L(1 + w + \varepsilon)} \leq \varepsilon + \frac{1}{L\varepsilon} = 3\varepsilon,$$

where we used $r\varepsilon \leq w \leq 1 + w + \varepsilon$. This single estimate covers all word lengths and both half-blocks, and proves (5).

Taking the average and difference now gives

$$\|f * \mu_n^* - f\|_\infty \leq 3 \cdot 2^{-n}, \tag{6}$$

$$\|d_n - 2\delta_e\|_\infty \leq 6 \cdot 2^{-n}, \qquad d_n = f * (\nu_n^* - (\nu_n')^*). \tag{7}$$

In particular,

$$\sum_{n \geq 1} \|d_{n+1} - d_n\|_\infty \leq 6 \sum_{n \geq 1} (2^{-(n+1)} + 2^{-n}) < \infty, \qquad d_n \longrightarrow 2\delta_e \neq 0.$$

This disproves the positive strong condition.

The endpoint estimates also give the geometric conclusion directly. The space X_f is separable because G is countable, and is invariant under the isometries R_g. Each probability average is a convex combination of right translates of f. Hence (5) places the two distinct limits $f + \delta_e, f - \delta_e$ in $\overline{\mathrm{co}}(\mathrm{Orb}_G(f))$. Their midpoint is f, which precludes an invariant strictly convex norm. □

5. Realizing the product conditions

It remains to produce a sequence satisfying the three hypotheses of Proposition 4.1 when $[G : \mathrm{FC}(G)] = \infty$. Throughout this section put $F = \mathrm{FC}(G)$. The group need not be countable.

5.1. Rewritability and the product conditions.

Finite index of the FC-center forces collisions among products. The following observation uses the prefix-coset argument in [BR, Lemmas 3.1–3.2].

Lemma 5.1. *If $[G : \mathrm{FC}(G)] < \infty$, every infinite sequence in G has two disjoint nonempty finite consecutive blocks whose products commute. In particular, its products of distinct-index words cannot all be distinct.*

Proof. Write the sequence as (s_i) and its prefix products as $p_k = s_1 \cdots s_k$, with $p_0 = e$. Two prefixes lie in the same coset of F, say $p_a F = p_b F$ with $a < b$. Hence the nonempty block product

$$u = p_a^{-1} p_b = s_{a+1} \cdots s_b$$

belongs to F. Now $C_G(u)$ has finite index. Apply the same pigeonhole argument to prefixes $q_k = s_{b+1} \cdots s_{b+k}$ of the remaining tail. Equality $q_c C_G(u) = q_d C_G(u)$, with $c < d$, gives

$$v = q_c^{-1} q_d = s_{b+c+1} \cdots s_{b+d} \in C_G(u).$$

Thus $uv = vu$, while the two corresponding concatenations of the disjoint blocks are different words with no repeated index. □

Blyth–Rhemtulla call a group *eventually rewritable*, or a $\mathcal{Q}_\infty$-group, if every sequence $(x_i)_{i \geq 1}$ admits $n \geq 2$ and distinct permutations σ, τ of $\{1, \ldots, n\}$ such that

$$x_{\sigma(1)} \cdots x_{\sigma(n)} = x_{\tau(1)} \cdots x_{\tau(n)}.$$

They prove that this is equivalent to being FC-by-finite, that is, virtually FC [BR, Proposition 2]. Consequently, outside this class there is a sequence whose products distinguish every ordering of each finite initial segment. They also distinguish orderings of any fixed finite subset: append its complement in an initial segment and cancel. This explains why products that retain their order are natural here. It does not yet distinguish different supports, exclude repeated appends, or control right successors outside the product set. Those are the extra requirements of Proposition 4.1.

Neither extra requirement follows merely from distinguishing ordered words. The standard generators of the free product $*_{i \geq 1} \mathbb{Z}_2$ distinguish all words with distinct indices, but $s_i^2 = e$ violates the repeated-append condition. Passing to a subsequence does not fix this.

The exterior-successor condition is also independent of unique decoding. For example, in the free group on $a, b, t, c_1, c_2, \ldots$, the sequence

$$a,\ b,\ ta,\ tb,\ c_1,\ c_2, \ldots$$

has distinct values for all positive words, even when indices repeat: read a symbol t together with the following a or b, and every other symbol separately. This uniquely decodes a product into the displayed generators, and also gives the repeated-append exclusion. Nevertheless, t is outside their positive product set and has the two successors ta, tb in it. Thus even unique decoding of all positive words would not by itself supply the analytic obstruction.

The basic induction is also suggested by the proof of [BR, Lemma 3.3]. If a new element x is inserted into two old products, a collision takes the form

$$axb = cxd \quad \Longleftrightarrow \quad x(bd^{-1})x^{-1} = a^{-1}c.$$

When $bd^{-1} \notin F$, this excludes at most one coset of an infinite-index centralizer. When $bd^{-1} \in F$, its centralizer has finite index; we instead choose x in that centralizer. The equation then reduces to an equality among old products. The induction must retain enough information to prove that each such surviving equality is harmless. The finite extension lemma below uses two necessary features of an infinite sequence with the desired predecessor property: nonempty products avoid F, and different letters lie in different cosets of F.

This use of centralizer cosets is shared with the switching literature. The ordinary switching argument is explicit in [FTVF, Claim 3.3]; super-switching adds equations of the form $xax = b$ in [FHTVF, Definition 2.3 and Proposition 2.5]. Erschler–Kaimanovich organize these arguments using negligible sets and prove super-switching existence for all ICC groups [EK, Sections 4.B–4.C, especially Proposition 4.25]. We use their notation below. Our group can have a nontrivial FC-center, and the common-predecessor requirement also produces $xax^{-1} = bxc$. We therefore

prove the precise simultaneous avoidance statement needed here, rather than assume it follows from a switching statement for ICC groups.

5.2. **Negligible sets and finite equation avoidance.** Following [EK, Definition 4.9], a subset of G is *negligible* if it is contained in a finite union of cosets of infinite-index subgroups. This family is closed under subsets, finite unions, and left and right translation. For right translation, use $aKb = ab(b^{-1}Kb)$.

Lemma 5.2 (Neumann's finite coset covering lemma [Neu])**.** *No finite-index subgroup of G is negligible.*

Proof. Neumann's theorem says that a group covered by finitely many cosets of subgroups has a finite-index subgroup among them; see also [EK, Lemma 4.13]. For the stated version, let $H \leq G$ have finite index. Each nonempty $H \cap aK$ is a coset in H of $H \cap K$. If $[G : K] = \infty$, then $[H : H \cap K] = \infty$. A negligible cover of H would therefore contradict the theorem applied to H. □

Following [EK, §§4.B–4.C], write the conjugacy and quadratic solution sets as

$$S_{a\to b} = \{x \in G : xax^{-1} = b\}, \qquad S^{\pm}_{a\to b} = \{x \in G : xax = b\}.$$

If $x_0 \in S_{a\to b}$, then

$$S_{a\to b} = x_0 C_G(a). \tag{8}$$

Indeed, $xax^{-1} = x_0 a x_0^{-1}$ if and only if $x_0^{-1}x$ commutes with a. Thus this solution set is negligible whenever $a \notin F$. Erschler–Kaimanovich also show that $S^{\pm}_{a\to b}$ is contained in a coset of $C_G(ab)$ [EK, Lemma 4.23]. This handles $ab \notin F$, but our induction must allow arbitrary coefficients, including the involution equation $x^2 = e$. The following overlap argument treats all the required quadratic and mixed equations at once.

Lemma 5.3 (Finite equation avoidance)**.** *Suppose $[G : F] = \infty$ and $H \leq G$ has finite index. No finite union of a negligible set and sets of the following two forms covers H:*

$$S^{\pm}_{a\to b} = \{x : xax = b\}, \qquad \{x : xax^{-1} = bxc\}, \qquad a, b, c \in G.$$

Proof. We first show that if E is either displayed solution set and $h \notin F$, then $E \cap h^{-1}E$ is negligible. Membership means $x, hx \in E$. In the quadratic case this gives

$$hxahx = xax \quad\Longrightarrow\quad hxah = xa \quad\Longrightarrow\quad x(aha^{-1})x^{-1} = h^{-1}.$$

Thus $E \cap h^{-1}E \subseteq S_{aha^{-1}\to h^{-1}}$. The element aha^{-1} is outside F, so (8) puts the intersection in a coset of an infinite-index centralizer, unless it is empty.

For the mixed equation $xax^{-1} = bxc$, substituting this relation into the equation for hx yields

$$hbxch^{-1} = bhxc \quad\Longrightarrow\quad x(ch^{-1}c^{-1})x^{-1} = b^{-1}h^{-1}bh.$$

Since $ch^{-1}c^{-1} \notin F$, the same reasoning proves negligibility of the intersection. This is a statement about the overlaps; it does not assert that E itself is negligible.

Suppose now that $H \subseteq A \cup E_1 \cup \cdots \cup E_r$, where A is negligible and the E_i are solution sets of the displayed forms. Since $[H : H \cap F] = \infty$, choose $h_0, \ldots, h_r \in H$ in distinct cosets modulo F. For each $x \in H$, either one of $h_j x$ belongs to A, or two of the $r + 1$ points belong to the same E_i. Hence

$$H \subseteq \bigcup_{j=0}^{r} h_j^{-1}A \;\cup \bigcup_{\substack{1\le i\le r\\ 0\le j<k\le r}} (h_j^{-1}E_i \cap h_k^{-1}E_i).$$

The pair intersection equals $h_j^{-1}(E_i \cap (h_k h_j^{-1})^{-1}E_i)$ and is negligible by the first part, since $h_k h_j^{-1} \notin F$. The right-hand side is thus negligible, contradicting Lemma 5.2. □

Related recursive avoidance arguments occur in [BP, Theorems 4 and 5], using centralizers and group Zariski topologies. No topological nondiscreteness assumption is needed in Lemma 5.3.

5.3. **The finite extension problem.** The three product requirements follow from a single condition, if we allow the identity as an empty terminal. Set $s_0 = e$ and use the convention $t0 = t$. The symbol 0 is never an entry of a word in $\mathcal{F}$; it is used only for the terminal indices below.

For a finite stage, let $\mathcal{F}_n$ be the words with entries in $\{1, \dots, n\}$ and $\mathcal{S}_n = \{s_v : v \in \mathcal{F}_n\}$. Consider the following *common-predecessor condition*:

$$\begin{aligned} v, w \in \mathcal{F}_n, \quad i, j \in \{0, \dots, n\}, \quad i \neq j, \quad s_v s_i^{-1} = s_w s_j^{-1} \\ \implies \quad v = ti, \quad w = tj \quad \text{for some } t \in \mathcal{F}_n. \end{aligned} \tag{9}$$

In words, if a point has right successors in the product set indexed by two distinct terminals, both must come from appending those terminals to one word. Allowing terminal 0 also handles a point already in the product set.

Condition (9) is an exact reformulation of the three product requirements, valid also at a finite stage. First, it implies each of them:

(1) If $s_v = s_w$ and $v = ti$ is nonempty, then $s_w s_i^{-1} = s_t s_0^{-1}$. The condition gives $w = ui$ and $t = u$, so $w = v$. Interchanging v, w if necessary proves injectivity; the case of two empty words is immediate.
(2) If $s_v s_i = s_w$ with $1 \leq i \leq n$, then $s_w s_i^{-1} = s_v s_0^{-1}$. The condition gives $w = vi$ as a word without repeated entries. Hence $i \notin \mathrm{ran}(v)$.
(3) If $g s_i = s_v$ and $g s_j = s_w$ with distinct $i, j \in \{1, \dots, n\}$, then the condition gives $v = ti$, $w = tj$, and therefore $g = s_t \in \mathcal{S}_n$.

Conversely, suppose the three requirements hold and put $g = s_v s_i^{-1} = s_w s_j^{-1}$. If one terminal is 0, then $g \in \mathcal{S}_n$ immediately. Otherwise, the exterior-successor condition gives the same conclusion. Write $g = s_t$. The repeated-append exclusion says that each nonzero terminal is absent from t, and injectivity now gives $v = ti$, $w = tj$. The same reasoning applies to an infinite sequence.

It remains to extend finite tuples while preserving (9). Two further conditions will be maintained:

$$s_v \notin F \quad (v \in \mathcal{F}_n \setminus \{\varnothing\}), \tag{10}$$

$$s_i F \neq s_j F \quad (1 \leq i < j \leq n). \tag{11}$$

Both conditions are necessary for any infinite sequence satisfying (9). For (10), suppose a nonempty word U has value $u \in F$. A nonempty block T using only later indices can be chosen with $s_T \in C_G(u)$, by the tail-prefix pigeonhole argument of Lemma 5.1. Then $s_{UT} = s_{TU}$ contradicts injectivity. For (11), if $h = s_i^{-1} s_j \in F$, choose such a later block T with $s_T \in C_G(h)$. Then

$$s_{jT} s_j^{-1} = s_j s_T s_j^{-1} = s_i s_T s_i^{-1} = s_{iT} s_i^{-1}.$$

Condition (9) would require the word jT to end in j, which it does not.

For a finite tuple these are additional hypotheses, since the tail used above is not available. In the extension proof they distinguish the possible insertion positions of the new letter.

Lemma 5.4 (Finite extension)**.** *Suppose $[G : F] = \infty$, $n \geq 0$, and a finite tuple $s_1, \dots, s_n$ satisfies (9), (10), and (11). There is an element $x \in G$ such that the augmented tuple $s_1, \dots, s_n, x$ satisfies all three conditions.*

Proof. Write $\star$ for the new formal letter $n + 1$ and x for its value. A new positive word has the form $A \star B$, where A, B are old words with disjoint sets of entries. Lowercase a, b, c, d denote the evaluations of old words A, B, C, D.

How conjugacy equations are treated. For each conjugacy equation

$$xux^{-1} = z \tag{12}$$

encountered below, if $u \notin F$ we exclude its solution set, which is empty or a coset of the infinite-index centralizer $C_G(u)$. For the finitely many coefficients $u \in F$, require $x \in C_G(u)$. Their centralizers have a finite-index intersection H; take $H = G$ if there are none. All coefficients are determined by the old sequence, so H is fixed before x is chosen. Within H, an equation (12) with $u \in F$ becomes the old equality $u = z$: the new letter has disappeared. We check below that these surviving equalities give precisely the required predecessor. All other forbidden sets are singletons, cosets of F, or the quadratic and mixed solution sets in Lemma 5.3.

Preserving the two FC-center conditions. A new positive product has the form axb, so (10) excludes the sets

$$\{x : axb \in F\} = a^{-1}Fb^{-1}.$$

Condition (11) also excludes $s_1F, \dots, s_nF$. Since F is normal and has infinite index, all of these are infinite-index cosets, and there are only finitely many of them.

Both distinguished terminals old. Take distinct $i, j \in \{0, \dots, n\}$ and consider $s_v s_i^{-1} = s_w s_j^{-1}$. If neither word contains $\star$, the old condition applies. If exactly one does, the equation fixes a single value of x. If both do, write $v = A \star B$, $w = C \star D$; then

$$axbs_i^{-1} = cxds_j^{-1} \quad \Longleftrightarrow \quad x(bs_i^{-1}s_jd^{-1})x^{-1} = a^{-1}c.$$

The choice rule leaves only $bs_i^{-1}s_jd^{-1} = a^{-1}c \in F$. Thus $(ab)s_i^{-1} = (cd)s_j^{-1}$, and the old predecessor condition gives $AB = Ti$ and $CD = Tj$ for an old word T.

Now A, C are prefixes of Ti, Tj, and $a^{-1}c \in F$. If one is a proper prefix of the other, $a^{-1}c$ is the value of a nonempty old word or the inverse of such a value, contradicting (10). If they are incomparable, they must be $A = Ti$, $C = Tj$, with $i, j \neq 0$. In this case $a^{-1}c = s_i^{-1}s_j \notin F$ by (11), again a contradiction. Hence $A = C$, and this common prefix lies in T. Inserting $\star$ at that position in T gives $\widetilde{T}$ with $v = \widetilde{T}i$ and $w = \widetilde{T}j$, as required. This also covers either terminal being 0.

One distinguished terminal new. By symmetry it remains to consider

$$s_v x^{-1} = s_w s_j^{-1}, \qquad j \in \{0, \dots, n\}. \tag{13}$$

There are four possibilities for the occurrence of $\star$ in v, w. If neither word contains it, the equation fixes x. If only $v = A \star B$ contains it and $w = C$, the equation is

$$xbx^{-1} = a^{-1}cs_j^{-1}.$$

If B is nonempty, (10) gives $b \notin F$, so this solution set is excluded. If B is empty, the equation is $as_j = c$. For $j \neq 0$, the old predecessor condition with terminals $0, j$ gives $C = Aj$. For $j = 0$, old injectivity, already deduced from that condition, gives $C = A$. Thus in either case $v = A\star$, $w = Aj$, exactly the required pattern.

If only $w = C \star D$ contains $\star$ and $v = A$, then

$$ax^{-1} = cxds_j^{-1} \quad \Longleftrightarrow \quad x(ds_j^{-1})x = c^{-1}a,$$

which is quadratic. If both words contain $\star$, then

$$axbx^{-1} = cxds_j^{-1} \quad \Longleftrightarrow \quad xbx^{-1} = a^{-1}c\,x\,(ds_j^{-1}),$$

which is mixed. Lemma 5.3 allows us to exclude both types.

Every possible failure of the three maintained conditions has now been accounted for. There are only finitely many equations at each stage, and singletons are negligible because G is infinite.

Applying Lemma 5.3 within H supplies the next element x. This proves the finite extension statement. □

Theorem 5.5. *If $[G : \mathrm{FC}(G)] = \infty$, there is a sequence $(s_i)_{i\geq 1}$ satisfying the three product hypotheses of Proposition 4.1. No countability assumption on G is needed for this construction.*

Proof. The empty tuple satisfies (9), (10), and (11) vacuously. Apply Lemma 5.4 successively. Any failure of (9) for the resulting infinite sequence would involve only finitely many entries and terminal indices, so would already occur at a finite stage. The three deductions above therefore give all the required product properties. □

6. The classification and examples

The preceding sections now give both implications of the main theorem.

Proof of Theorem 1.1. Conditions (2) and (3) are equivalent by [DD, Theorem 4.2]. Corollary 3.3 proves (1)⇒(2), and restricting to probability measures gives (3)⇒(4).

If (1) fails, (2) gives $[G : \mathrm{FC}(G)] = \infty$. Theorem 5.5 supplies the sequence required by Proposition 4.1, whose probability measures violate (4). This proves (4)⇒(1) and completes the equivalence. □

6.1. **Permanence and examples.** In particular, among countable groups the strongly Choquet–Deny groups are exactly the eventually rewritable ($\mathcal{Q}_\infty$) groups, by [BR, Proposition 2] and Theorem 1.1. The following consequences distinguish the classification from both virtual abelianness and the ordinary Choquet–Deny property.

Corollary 6.1. *Among countable groups, the strongly Choquet–Deny property is preserved by subgroups, quotients, finite direct products, and passage to a group containing the given group with finite index.*

Proof. These are permanence properties of virtually FC groups. If $H \leq G$ is FC of finite index, then $K \cap H$ is FC of finite index in every subgroup $K \leq G$, and the image of H in any quotient is FC of finite index. Products and finite-index overgroups are immediate. Apply Theorem 1.1. □

The subgroup and finite-index permanence were already established in [DD, Proposition 5.1]. The classification also makes clear why increasing unions need not preserve the property.

Example 6.2. Every countable direct sum of finite groups is strongly Choquet–Deny: conjugation preserves the finite support of each element, so the group is FC. More generally, a countable increasing union of finite normal subgroups is FC, since the conjugacy class of each element stays in one of those subgroups.

In particular, $G_0 = \bigoplus_{n\geq 1} S_3$ is strongly Choquet–Deny but is not virtually abelian. To see the latter assertion, let $F_n = \prod_{j=1}^{n} S_3$ and let $A \leq G_0$ be abelian. Each coordinate projection of $A \cap F_n$ has order at most three, so

$$|A \cap F_n| \leq 3^n, \qquad [G_0 : A] \geq [F_n : A \cap F_n] \geq 2^n.$$

Thus A has infinite index.

Example 6.3. Let $D_\infty = \mathbb{Z} \rtimes (\mathbb{Z}/2\mathbb{Z})$, where the nontrivial element acts by inversion, and put $G = \bigoplus_{n\geq 1} D_\infty$. The FC-center of D_∞ is its translation subgroup $\mathbb{Z}$: a translation has at most two conjugates, whereas every reflection has infinitely many. Conjugation in G is coordinatewise, and hence

$$\mathrm{FC}(G) = \bigoplus_{n\geq 1} \mathbb{Z}, \qquad G/\,\mathrm{FC}(G) \cong \bigoplus_{n\geq 1} \mathbb{Z}/2\mathbb{Z}.$$

The FC-center has infinite index, so G is not strongly Choquet–Deny. Nevertheless, G is an increasing union of the normal direct factors $\prod_{j=1}^{n} D_\infty$, each of which is virtually abelian and strongly Choquet–Deny. Thus neither increasing unions of normal strongly Choquet–Deny subgroups nor countable direct sums preserve the property.

This group is ordinary Choquet–Deny. Indeed, in an ICC quotient the image of $\mathrm{FC}(G)$ must be trivial, because its elements retain finite conjugacy classes. Such a quotient would then be abelian, hence trivial. The assertion follows from [FHTVF, Theorem 1].

Example 6.4. The integral Heisenberg obstruction is [DD, Example 5.5]. The following locally finite variant illustrates why the full countable classification requires more than the finitely generated case. Let p be a prime and $R = \mathbb{F}_p[t]$. The Heisenberg group

$$H_3(R) = \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in R \right\}$$

is countable, locally finite, and nilpotent of class two. For local finiteness, the first and second coordinates of a finitely generated subgroup lie in finite-dimensional $\mathbb{F}_p$-spaces, and its third coordinates lie in the span of the original third coordinates and the pairwise products of the first and second coordinates.

Its FC-center is exactly its center, consisting of the elements with $a = b = 0$. Indeed, if $a \neq 0$, conjugation by the elements with coordinates $(0, b', 0)$ gives infinitely many distinct shifts ab' in the central coordinate. If $b \neq 0$, the analogous conjugations by $(a', 0, 0)$ give the same conclusion. We use here that R is an infinite integral domain. Thus $[H_3(R) : \mathrm{FC}(H_3(R))] = \infty$, and the group is not strongly Choquet–Deny.

Every nontrivial quotient is nilpotent of class at most two and has nontrivial center, so no quotient is ICC. It follows again from [FHTVF, Theorem 1] that $H_3(R)$ is ordinary Choquet–Deny. This gives a locally finite example separating the two properties.

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